

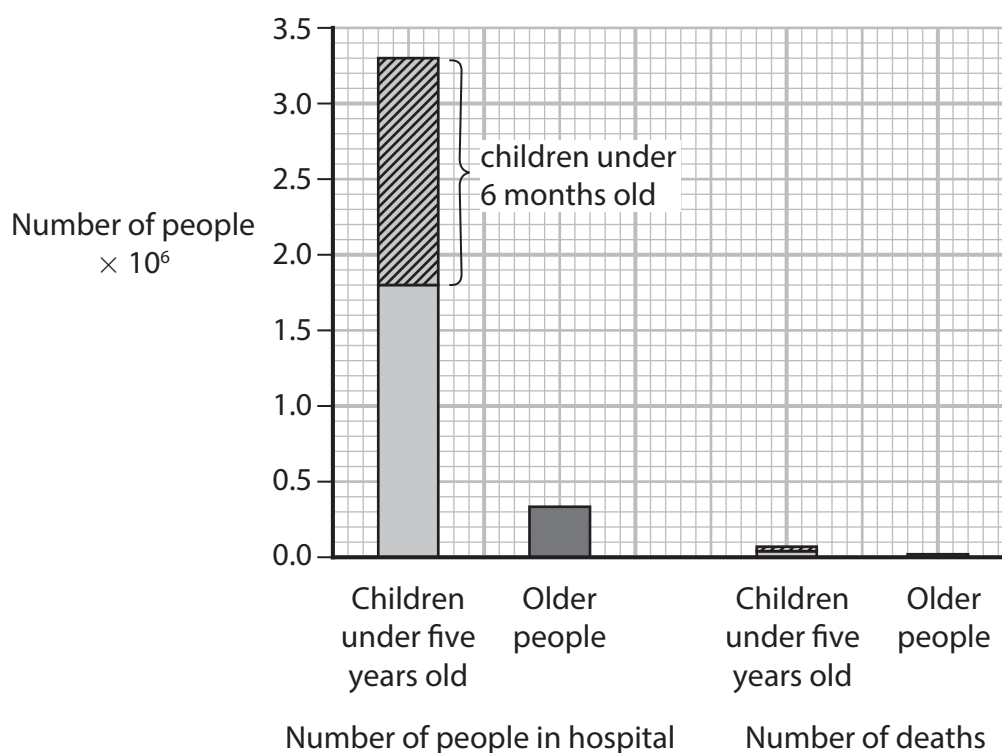
- 6 Respiratory syncytial virus (RSV) usually causes mild, cold-like symptoms from which most people recover.

Infection with RSV can be serious in young children and older people.

Currently, there are no approved vaccines against RSV, although there are some in clinical trials.

- (a) The graph shows the number of people who had to go to hospital in one country in one year.

The graph also shows the number of deaths in children under five years old and in older people in the same country in that year.



- (i) Which is the proportion of the children under five in hospital who are under 6 months old?

(1)

☐ **A** $\frac{5}{11}$

☐ **B** $\frac{6}{11}$

☐ **C** $\frac{5}{6}$

☐ **D** $\frac{6}{5}$

(ii) Which would be a more appropriate scale to use on the y-axis so that the range of values for the number of deaths can be seen clearly?

(1)

- ☐ **A** linear
- ☐ **B** logarithmic
- ☐ **C** standard deviation
- ☐ **D** standard form

(b) In one trial, people received a new vaccine against RSV.

The results showed that this vaccine protected against two subtypes of RSV.

Give **two** reasons why one vaccine can protect against two subtypes of RSV.

(2)

(c) In another trial, pregnant women were given a vaccine against RSV.

This vaccine did not cross from the mother into the developing fetus.

(i) As a result of passive immunity, the babies of these pregnant women were born with high levels of antibodies against RSV.

Explain how passive immunity developed in these babies before they were born.

(5)

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- (ii) The scientists running this trial claimed that the vaccine was 85 % successful at protecting babies against RSV infections that required medical attention.

Describe how the data could have been collected to support this claim.

(3)

(Total for Question 6 = 12 marks)
